

JINGHUAN SHANG

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Employment

Robotics and AI Institute, Boston, MA, USA

Sep 2024 – Present

Research Scientist

Generative simulation, foundation models, and robot learning

Education

Stony Brook University, NY, USA

2018 – 2024

Ph.D. in Computer Science, GPA: 3.98/4, Advisor: Prof. Michael S. Ryoo

Shanghai Jiao Tong University, Shanghai, China

2014 – 2018

B.S. in Computer Science, IEEE Pilot Class

Interests

Robot learning, automated systems for robot learning, speeding up anything, training, fine-tuning, or applying foundation models for robotics, simulation.

Selected Publications

1. **Shang, J.**, Patel, H., Gong, R. & Schmeckpeper, K. Sceniris: A Fast Procedural Scene Generation Framework (2025).
2. Gong, Ran*, Zhang, Xiaohan*, **Shang, Jinghuan***, Minniti, Maria Vittoria* & et al. ANYTASK: an Automated Task and Data Generation Framework for Advancing Sim-to-Real Policy Learning (2025).
3. Wang, Y. *et al.* VER: Vision Expert Transformer for Robot Learning via Foundation Distillation and Dynamic Routing in *International Conference on Learning Representations (ICLR)* (2026).
4. Li, X. *et al.* LLaRA: Supercharging Robot Learning Data for Vision-Language Policy in *International Conference on Learning Representations (ICLR)* (2025).
5. **Shang, J.** *et al.* Theia: Distilling Diverse Vision Foundation Models for Robot Learning in *Conference on Robot Learning (CoRL)* (2024).
6. Bai, X., **Shang, J.**, Sun, Y. & Balasubramanian, N. *Continual Learning with Global Alignment* in *Proceedings of Conference on Neural Information Processing Systems (NeurIPS)* (2024).
7. Li, X., Belagali, V., **Shang, J.** & Ryoo, M. S. Crossway Diffusion: Improving Diffusion-based Visuomotor Policy via Self-supervised Learning in *IEEE International Conference on Robotics and Automation (ICRA)* (2024).
8. **Shang, J.** & Ryoo, M. S. *Active Reinforcement Learning under Limited Visual Observability* in *Proceedings of Conference on Neural Information Processing Systems (NeurIPS)* (2023).
9. **Shang, J.**, Das, S. & Ryoo, M. S. *Learning Viewpoint-Agnostic Visual Representations by Recovering Tokens in 3D Space* in *Proceedings of Conference on Neural Information Processing Systems (NeurIPS)* (2022).

10. Li, X., **Shang, J.**, Das, S. & Ryoo, M. S. *Does Self-supervised Learning Really Improve Reinforcement Learning from Pixels?* in *Proceedings of Conference on Neural Information Processing Systems (NeurIPS)* (2022).
11. Burgert, R., **Shang, J.**, Li, X. & Ryoo, M. S. *Neural Neural Textures Make Sim2Real Consistent* in *Conference on Robot Learning (CoRL)* (2022).
12. **Shang, J.**, Li, X., Kahatapitiya, K., Lee, Y.-C. & Ryoo, M. S. StARformer: Transformer with State-Action-Reward Representations for Robot Learning. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)* (2022).
13. **Shang, J.**, Kahatapitiya, K., Li, X. & Ryoo, M. S. *StARformer: Transformer with State-Action-Reward Representations for Visual Reinforcement Learning* in *European Conference on Computer Vision (ECCV)* (2022).
14. **Shang, J.** & Ryoo, M. S. *Self-Supervised Disentangled Representation Learning for Third-Person Imitation Learning* in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (2021).

Research Experience

Research Scientist, Robotics and AI Institute
Generative simulation and sim2real [1, 2]

Sep 2024 - Present

- Automated pipeline for generating simulation tasks and policies from language description and assets.
- Automated pipeline for labeling asset attributes in natural language and retrieval. Processed **100k assets**.
- High-performance GPU-accelerated motion planning (**1000x faster**) and **collision-free, robot-reachable** scene generation (**200x faster**). **10k scene generation in 1min**.
- Embodiment-agnostic system designs for easy adaptation to new robots in sim.
- Leading role in implementing and grounding the entire system with Isaac Sim.

Robot Learning

- Action representation for robot policy that changes its execution speed. **2x speed** with minor success rate degradation. **Faster than teleop demonstration**.
- Efficient data generation for world-action-model finetuning.

Research Intern, Boston Dynamics AI Institute
Foundation models for robot learning [3, 5]

Jan 2024 – Jun 2024

- Distill multiple vision foundation models for robot learning. Large-scale training and data processing (400+ TB)
- 1.4M ImageNet pretraining beats 100M+ image pretraining on robot tasks. **10% task improvement with only 1% of the compute**.
- Mixture-of-expert structure for vision representation in robot learning.

Research Intern, Motional AD Inc
 Trajectory prediction models, benchmarks and datasets

Aug 2022 – Dec 2022

Research Assistant, Stony Brook University
Visual Representation Learning for Robotics

May 2020 – Sep 2024

- **Active Vision Reinforcement Learning [8]**
 - * Learning to see and act simultaneously through a task in limited observable environments.

- **Viewpoint-agnostic Representation** [9]
 - * A learnable, differentiable, and light-weighted plugin for Transformer that learns viewpoint-agnostic representations from monocular 2D image.
- **Imitation Learning for Egocentric Tasks from Third-person Experiences** [14]
 - * A disentanglement approach to align first-person view and third-person view experiences for reward assignment that benefits the robot learning from third-person view observations.

Sequence Representation Learning for Robotics

- **Transformer for Robot Learning** [12, 13]
 - * StARformer, a decision transformer model with explicit separate local and long-horizon representations for better offline-RL and imitation performance, especially for longer sequences.
- **Diffusion Model for Sequential Behavior Generation** [7]
 - * Crossway Diffusion, a diffusion model with a self-supervised branch that enhances imitation learning.

Professional Activities

Conference Organizer: Web Chair, CoRL 2025

Conference Area Chair: CoRL 2026

Conference Reviewer: CVPR'22,'24,'25, ECCV'22, AISTATS'23,'25, ICML'23,'24,'25, ICCV'23,'25, NeurIPS'23,'24, AAAI'24, ICLR'22,'24,'25, ICRA'24

Guest Talk: Google Inc. (2022, Transformer for Robot Learning), CSE527 Introduction to Computer Vision (Fall 2021, graduate level), CSE525 Introduction to Robotics (Spring 2023, 2022, 2021, graduate level)

Teaching Assistant: CSE378 Introduction to Robotics (Fall 2023), CSE548 Analysis of Algorithms (Spring 2019, graduate level), CSE564 Visualization (Spring 2020, graduate level), CSE101 Computer Science Principles (Fall 2018)

Honors and Awards

NeurIPS 2022 Scholar Award	2022
Chairman's Fellowship, Stony Brook University	2018-2019
Outstanding Graduate of Colleges and Universities in Shanghai, China (Top 5%)	2018
1st Prize in China Undergraduate Mathematical Contest in Modeling	2017
Academic Excellence Scholarship of SJTU (Top 20%)	2015, 2016, 2017

Technical Skills

Competitive Programming: [\[My LeetCode\]](#) Ranked 9/54 in SBU ACM ICBC Selection Contest, 2020

Technologies/Frameworks: PyTorch, Linux, Git, Tensorflow, Unity3D

Simulated Environments: Isaac Sim, Pybullet, MuJoCo, DeepMind Control Suite, Robosuite, Atari.py

3D Assets: OpenUSD, Blender, Open3D